

# Zuzanna Stachlewska

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/staˈxlɛfska/

## Personal profile

Materials engineer with over 3 years of experience as materials researcher and a Master's degree in Materials and Manufacturing Engineering from DTU, driven by sustainable, well-designed and high-quality engineering solutions. I specialize in surface engineering and materials investigation, working with SEM/EDS, XRD, GD-OES, LOM, metallography and mechanical testing. Strong science communicator who enjoys tailoring complexity to the audience. Analytical, detail-oriented and proudly a transgender woman who values inclusivity and good collaboration. Outside the lab, you'll find me cycling through North Zealand's forests and hills or capturing details of the surrounding world with a camera in hand.

## Experience **Materials and Surface Specialist - Expanite A/S**

**2023-01 – Present**

### Achievements:

Developed and delivered to the market a novel surface hardening solution with potential multi-million DKK annual revenue. <sup>[1,2,3,4,6,8,10,13,14]</sup>

Developed a surface hardening solution for an automotive customer, currently in ramp-up phase toward high-volume production. <sup>[1,2,3,4,6,8,10,13,14]</sup>

Reduced process times by 5% through optimisation of furnace cycles. <sup>[4,6,10]</sup>

### Key responsibilities:

**Leading customer projects** <sup>[13]</sup> in surface hardening or heat treatment, including vendor coordination.

**Performing R&D** <sup>[1,2,3,4,6,8,10,13,14]</sup> activities, including qualitative and quantitative laboratory analyses.

**Conducting metallurgical investigations** <sup>[1,2,3,6]</sup>, including sample preparation, light optical microscopy (LOM), and hardness analysis.

**Analyzing results** <sup>[6,11,14]</sup> and communicating findings to internal and external stakeholders.

**Planning, executing, and evaluating tests** <sup>[1,6,8,11,13]</sup> for core products and processes.

**Troubleshooting** <sup>[6,10,14]</sup> vacuum systems (furnaces) and analyzing heat treatment process data.

**Investigating non-conformities** <sup>[14]</sup> and customer claims using root cause analysis (RCA) methodologies.

**Providing technical support** <sup>[3,6,11]</sup> to Production, Sales, and Quality teams.

**Designing and implementing** <sup>[6,7,8,13]</sup> technical fixtures and vacuum analysis auxiliary systems.

**Managing and creating technical documentation:** <sup>[10,11]</sup> PPAP, SOP, and QMS documentation.

**Writing** <sup>[11]</sup> whitepapers and creating technical materials for marketing team.

## Information

### Pronouns:

She/Her

### E-mail:

Please contact me via LinkedIn for the mail adress

### LinkedIn:

/zuzanna-stachlewska

### Telephone number:

Please contact me via LinkedIn for the telephone number

### Date of birth:

1997-05

### Address:

8240 Risskov

Relocation to Greater Copenhagen Area is possible

## Hard skills

- <sup>[1]</sup> Materials characterisation
- <sup>[2]</sup> Surface engineering
- <sup>[3]</sup> Metallurgy
- <sup>[4]</sup> Heat treatment
- <sup>[5]</sup> Additive manufacturing
- <sup>[6]</sup> Data analysis
- <sup>[7]</sup> Technical design (CAD)
- <sup>[8]</sup> Experiment design
- <sup>[9]</sup> PVD hard coatings

## Soft skills

- <sup>[10]</sup> Problem-solving
- <sup>[11]</sup> Communication
  - Communicating complex ideas to various stakeholders with tailored level of complexity
- <sup>[12]</sup> Life-long learner
- <sup>[13]</sup> Project management
  - Managing multiple projects across organisation
- <sup>[14]</sup> Analytical skills
  - Applying RCA methodologies to solve QA tasks and equipment malfunctions

2020–09 to 2022–08

### Technical University of Denmark (DTU) – Faculty of Mechanical Engineering

*MSc in Materials and Manufacturing Engineering*

**Thesis:** Additive manufacturing of lead-free (KNa)NbO<sub>3</sub> piezoelectric ceramics with the feasibility study of a templated growth approach. <sup>[1,5,6,7,8]</sup>

**Best project award:** Design of hearing aid battery compartment utilizing AM fast prototyping approach (Micro product design, development and production - DTU: 41743). <sup>[5,7,10]</sup>

2016–09 to 2020–07

### Lodz University of Technology (TUL) – Faculty of Mechanical Engineering

*BSc Eng. in Materials Engineering*

**Thesis:** Metal-Ceramic Composite in a SiC/Inconel 617 system as a potentially new material for brake systems. <sup>[1,2,3,4,6,8,9]</sup>

**Scholarship:** Awarded with a scholarship granted to 10% of best ranking students.

2019–02 to 2019–07

### Università degli studi di Napoli Federico II (UNINA) – Dipartimento di Ingegneria Chimica, dei Materiali e della Produzione Industriale (Erasmus+)

*MSc level courses during BSc Eng. programme*

*Courses conducted entirely in Italian.*

## Software skills

### Data Analysis <sup>[6]</sup>

OriginLab / Fiji (ImageJ) / Excel



### Technical writing tools <sup>[11]</sup>

LaTeX (Overleaf) / Typst



### Microsoft Office package

Excel / Word / PowerPoint



### Computer Aided Design (CAD) <sup>[7]</sup>

FreeCAD / SOLIDWORKS



### Programming <sup>[12]</sup>

Python (GUI app, GitHub repository [\[Link\]](#)) / Matlab



### Visual communication <sup>[11]</sup>

Adobe Illustrator / Adobe Photoshop / Inkscape / Affinity



## English

Full professional proficiency  
(8.0 IELTS score, 2020)



## Polish

Native



## Danish

Intermediate  
(PD2 certificate, 2023)



## Italian

Intermediate

